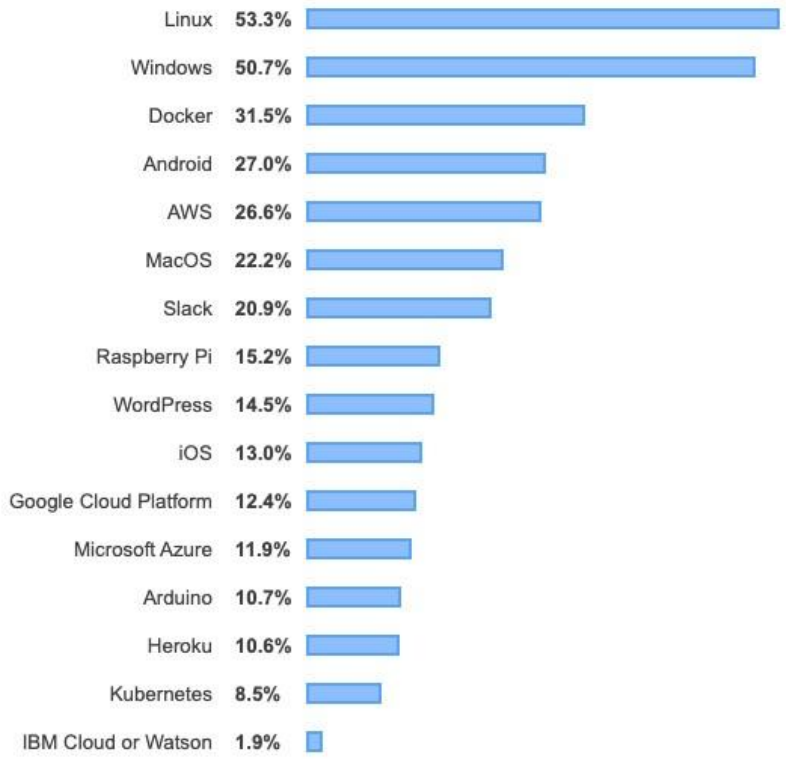
Linux Fundamentals for DevOps

Why Linux?

Platforms

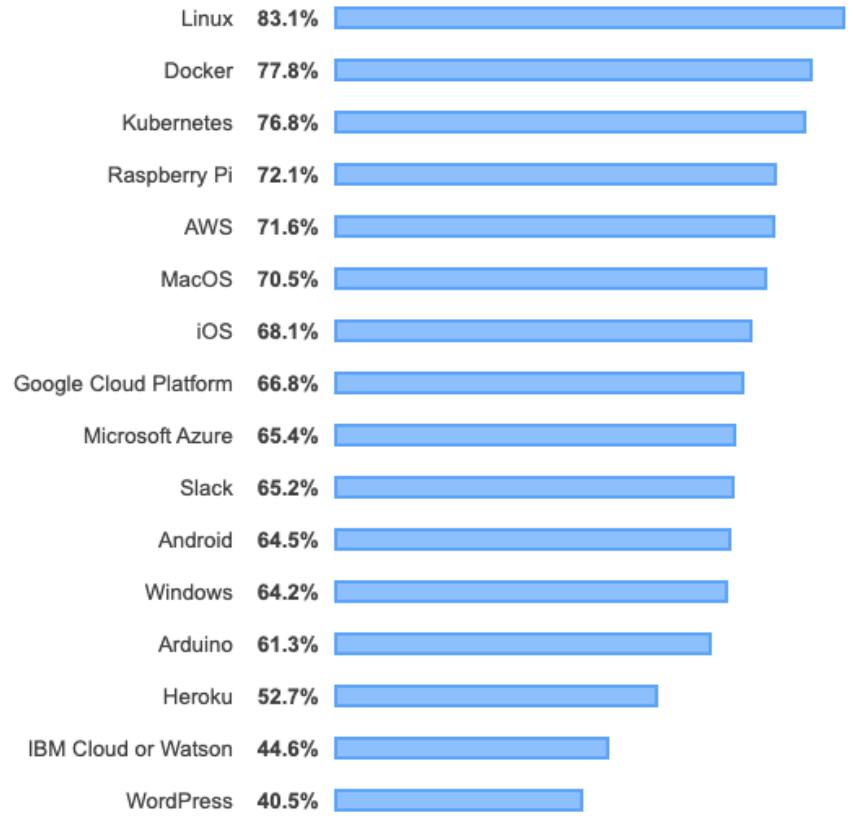
All Respondents | Professional Developers



80,144 responses; select all that apply

Most Loved, Dreaded, and Wanted Platforms

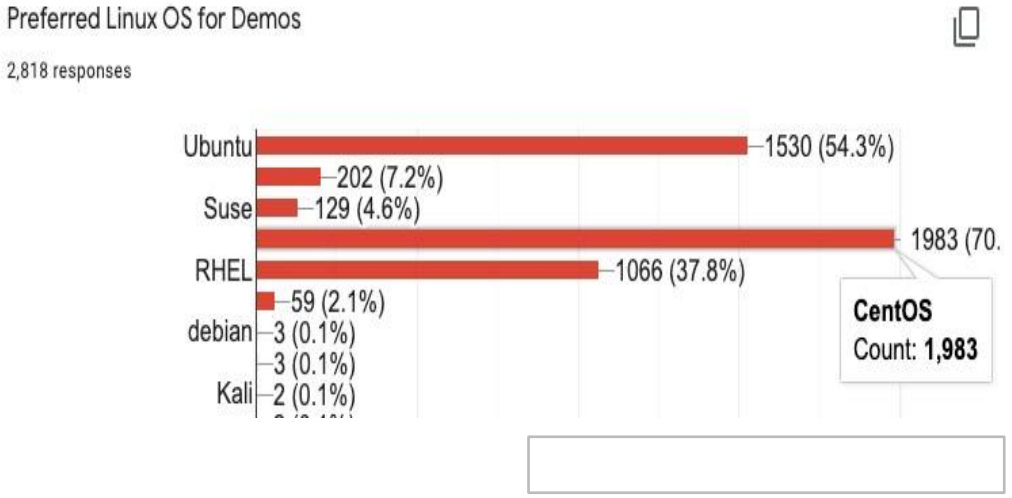
Loved | Dreaded | Wanted



https://insights.stackoverflow.com/survey/2019

Linux Basics

- Linux CLI
- VI Editor
- Package Management



Shell Types

```
echo $SHELL
/bin/bash
```

- Broune Shell (Sh Shell)
- C Shell (csh ortcsh)
- Z Shell (zsh)
- Bourne again Shell (bash)

Basic Commands

```
echo Hi
```
```
Hi
```
Print to screen

```
ls
```
```
File.txt my_dir1 file2.conf
```
List files & folders

```
cd my_dir1
```
Change directory

```
pwd
```
```
/home/my_dir1
```
Present Working Directory

```
mkdir new_directory
```
Make Directory

```
cd new_directory; mkdir www; pwd
```
```
/home/my_dir1/new_directory
```
Multiple commands

/tmp/asia/india/bangalore

```
mkdir  /tmp/asia
mkdir  /tmp/asia/india
mkdir  /tmp/asia/india/bangalore
```



```
mkdir -p /tmp/asia/india/bangalore
```

Make DirectoryHierarchy

```
rm -r /tmp/my_dir1
```

Remove Directory

```
cp -r my_dir1 /tmp/my_dir1
```

Copy Directory

Commands - Files

```
touch new_file.txt
Hi
```

Create a new file (no contents)

```
cat > new_file.txt
This is some sample contents
CTLR + D
```

Add contents tofile

```
cat new_file.txt
This is some sample contents
```

View contents offile

```
cp new_file.txt copy_file.txt
```

Copy File

```
mv new_file.txt sample_file.txt
```

Move (Rename) File

```
rm new_file.txt
```

Remove (Delete) File

Paths & Wildcards

Absolute path starts from /, for example /var/log/app.log.

Relative path starts from the current directory, for example ./app.log.

. means current directory; .. means parent directory; ~ means home directory.

* matches any sequence of characters.

? matches one character.

Use quotes when filenames contain spaces or shell metacharacters.

```
ls *.log
ls file?.txt
cp ./config/*.conf /tmp/backup/
cat 'project notes.txt'
```

Pipes & Redirection — Why They Matter

Linux commands normally read from standard input (stdin) and write to standard output (stdout).

Standard error (stderr) is used for error messages.

| sends stdout of one command to stdin of another command.

> writes stdout to a file and replaces existing content.

>> appends stdout to a file.

< supplies a file as stdin.

2> redirects stderr; 2>&1 combines stderr with stdout.

Pipes — Chaining Commands

```
# Count running processes containing "python"
ps aux | grep python

# Find the 10 largest files/lines after sorting
du -ah . | sort -h | tail -10

# Count matching log entries
grep "ERROR" app.log | wc -l

# Extract a field from command output
ps aux | awk '{print $1}' | sort | uniq
```

Redirection — stdin, stdout and stderr

Operator	Examples
> file — overwrite file	ls -l > files.txt
>> file — append to file	echo 'Build started' >> build.log
< file — read stdin from file	sort < names.txt
2> errors.log — redirect stderr	find / -name '*.conf' 2> errors.log
2>> errors.log — append stderr	command &> all-output.log
&> output.log — redirect stdout + stderr (bash)	

Pipes & Redirects — DevOps Scenario

Scenario: a web application writes hundreds of log entries.

```
grep -i 'error' app.log | tee error-only.log | wc -l
 # tee displays output and also writes it to a file
```

User Accounts

```
whoami
```

```
matthew
```

```
id
```

```
uid=1001(matthew) gid=1001(matthew) groups=1001(matthew)
```

```
su aparna
```

```
Password:
```

```
ssh aparna@192.168.1.2
```

User Accounts

```
ls /root
ls: cannot open directory /root: Permission denied
```

```
sudo ls /root
anaconda-ks.cfg initial-setup-ks.cfg
```

SUDO
/etc/sudoers

Download Files

```
curl http://www.some-site.com/some-file.txt -O
```

```
some-file.txt
```

```
wget http://www.some-site.com/some-file.txt -O some-file.txt
```

```
some-file.txt
```

Check OS Version

```
ls /etc/*release*
```

```
/etc/centos-release          /etc/os-release      /etc/system-release
/etc/centos-release-upstream /etc/redhat-release /etc/system-release-cpe
```

```
cat /etc/*release*
```

```
CentOS Linux release 7.7.1908 (Core)
Derived from Red Hat Enterprise Linux 7.7 (Source)
NAME="CentOS Linux"
VERSION="7 (Core)"
ID="centos"
ID_LIKE="rhel fedora"
VERSION_ID="7"
PRETTY_NAME="CentOS Linux 7 (Core)"
ANSI_COLOR="0;31"
CPE_NAME="cpe:/o:centos:centos:7"
HOME_URL="https://www.centos.org/"
BUG_REPORT_URL="https://bugs.centos.org/"
```

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Linux Package Managers

A package manager installs software and resolves dependencies.

RPM-based systems commonly use dnf or yum; rpm works directly with .rpm packages.

Debian/Ubuntu systems commonly use apt; dpkg works directly with .deb packages.

Package managers also provide search, update, upgrade and removal operations.

Prefer repositories and package managers over manually downloading binaries.

RPM / DNF / YUM — Red Hat Family

Modern commands

dnf search nginx

dnf install nginx

dnf update

dnf remove nginx

dnf list installed

dnf info nginx

Compatibility / low-level

yum install nginx

yum update

yum remove nginx

rpm -qa — list installed RPMs

rpm -qi nginx — package information

rpm -ql nginx — files installed by package

APT / DPKG — Debian & Ubuntu

APT

sudo apt update — refresh package index

sudo apt upgrade — upgrade installed packages

sudo apt install nginx

sudo apt remove nginx

apt search nginx

apt show nginx

DPKG

dpkg -l — list installed packages

dpkg -s nginx — package status

dpkg -L nginx — files installed

sudo dpkg -i package.deb

sudo apt -f install — repair dependencies

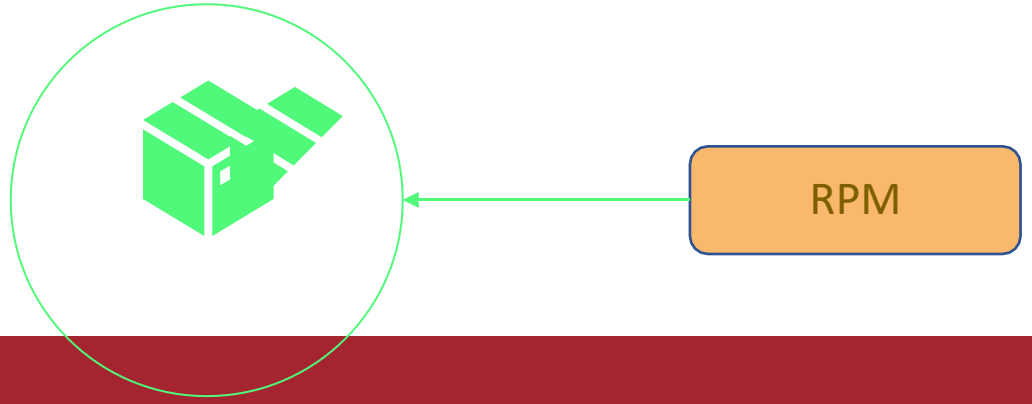
RPM (Red Hat Package Manager)

```
rpm -i telnet.rpm

rpm -e telnet.rpm

rpm -q telnet.rpm
```

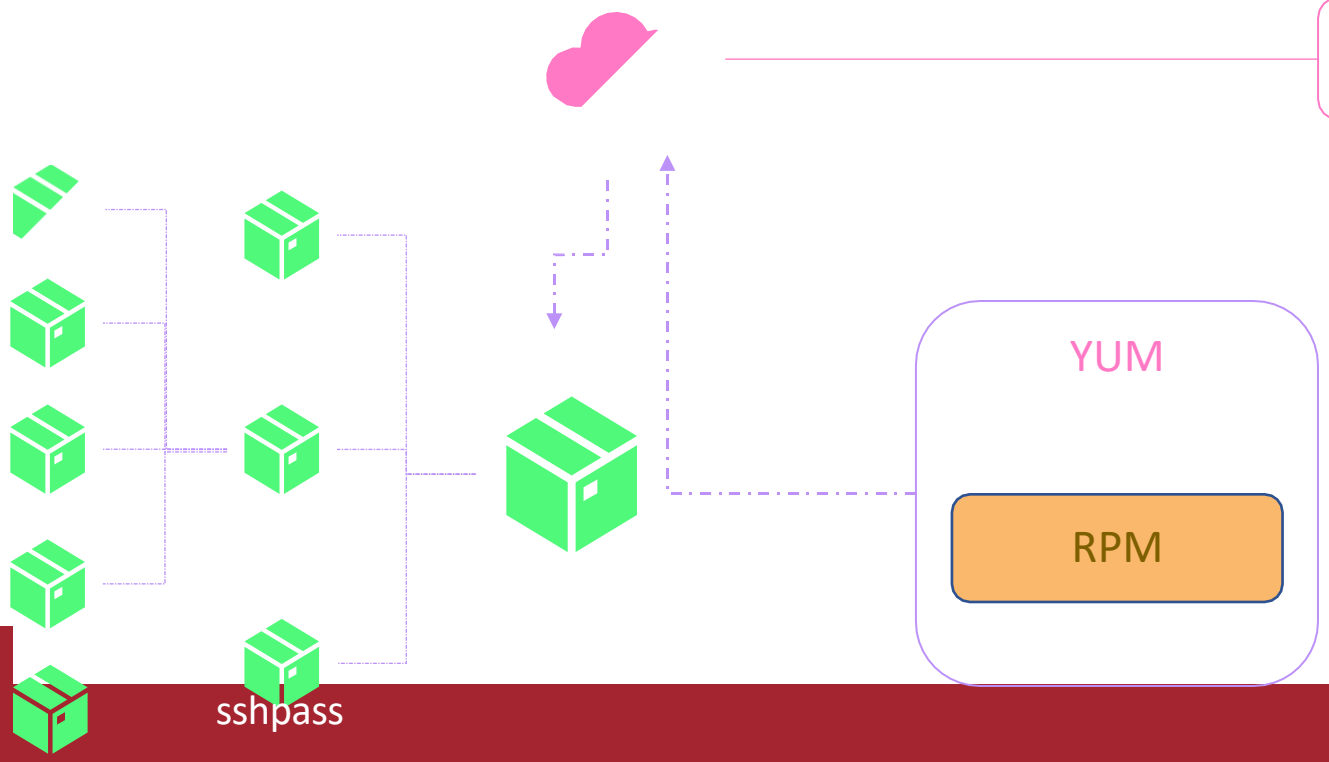
- Install Package
- Uninstall Package
- Query Package



```
yum install ansible
```

Install Package

/etc/yum.repos.d



YUM Repos

Extra Packages for Enterprise Linux (EPEL)

Welcome to the home of the EPEL Special Interest Group.

Quickstart

- epel-release-latest-6
- epel-release-latest-7
- epel-release-latest-8

You may retrieve signed binary configuration files from one the above two links (varying by the major release nu
automatically installed by root thus:

- RHEL/CentOS 6:

```
# yum install https://dl.fedoraproject.org/pub/epel/epel-release-latest-6.noarch.rpm
```

- RHEL/CentOS 7:

```
# yum install https://dl.fedoraproject.org/pub/epel/epel-release-latest-7.noarch.rpm
```

Vim — Essential Commands

Editing

i — insert before cursor

a — insert after cursor

o — new line below

dd — delete line

yy — copy line

p — paste

Save / Exit / Search

:w — save

:q — quit

:wq — save and quit

:q! — quit without saving

/pattern — search

n / N — next / previous match

Vim — Practical Editing Exercise

Create a configuration file: vim app.conf

```
vim app.conf
# Insert mode: i
# Save: Esc → :w
# Search: Esc → /PORT
# Save & exit: Esc → :wq
```